Acoustic Digital Twin and Adaptive Control for UAV Noise Mitigation: an LMS/FxLMS Benchmark Framework

[Author Name], *Student Member, IEEE*

Abstract

Unmanned Aerial Vehicles (UAVs) generate structured acoustic noise dominated by the Blade Passing Frequency (BPF) and its harmonics, posing challenges for urban deployment and noise-sensitive operations. This work presents a *simulation-first* framework for Active Noise Control (ANC) of UAV propeller noise using adaptive filtering. A BPF-based synthetic signal model is first established to parameterize UAV acoustic signatures from rotor geometry. Two adaptive algorithms are evaluated: the Least Mean Squares (LMS) filter as a validated baseline, and the Filtered-x LMS (FxLMS) algorithm incorporating a secondary acoustic path model to capture realistic loudspeaker–room dynamics. Experimental results show that LMS achieves a noise reduction of $24.53\,\mathrm{dB}$ at the optimal adaptation rate ($\mu = 0.01$), with clear suppression of the $200\,\mathrm{Hz}$ BPF and the $400\,\mathrm{Hz}$ second harmonic. FxLMS, operating with a 32-tap secondary-path impulse response, achieves $22.83\,\mathrm{dB}$ reduction—a $1.70\,\mathrm{dB}$ trade-off representing the cost of acoustic robustness. The paired comparison provides a reproducible digital-twin benchmark for ANC algorithm evaluation in UAV contexts, suitable as a foundation for real-time and spatially-extended deployments.

Index Terms

Active Noise Control, UAV acoustics, LMS adaptive filter, FxLMS, secondary path estimation, Blade Passing Frequency, digital twin.

I. Introduction

The rapid proliferation of UAVs in urban environments has intensified concerns about the acoustic impact of rotor noise. Unlike broadband machinery noise, UAV propeller noise is *tonally dominated*: discrete spectral peaks arise at the Blade Passing Frequency and its integer multiples, making it both perceptually salient and, in principle, amenable to feedforward active cancellation [2], [4].

Existing ANC literature focuses primarily on HVAC duct noise, headphones, and vehicle cabins. UAV-specific adaptation—accounting for time-varying RPM, multi-rotor interference, and the absence of a fixed secondary-path—remains comparatively under-studied [7], [8].

This paper makes three contributions:

1) A physics-derived, parametric UAV noise model linking rotor geometry (blade count, RPM) to a deterministic BPF signal with additive broadband turbulence.
2) A reproducible simulation pipeline—implemented in Python and validated on synthetic data—that exercises LMS and FxLMS under identical conditions, enabling a controlled comparison.
3) A quantitative performance benchmark ($24.53\,\mathrm{dB}$ for LMS, $22.83\,\mathrm{dB}$ for FxLMS) that exposes the robustness cost of secondary-path modeling and establishes a reference point for future hardware or spatially-extended studies.

Section II reviews UAV acoustics and adaptive ANC; Section III describes the signal model and algorithms; Section IV presents quantitative results; Section V interprets the findings; and Section VI summarises directions for future work.

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Author is with [University/Research Center], [City], [Country] (e-mail: email@institution.edu).

II. Background

A. UAV Acoustic Signature

A B-blade rotor spinning at Ω revolutions per minute produces a fundamental tone at the Blade Passing Frequency:

$$f_{\text{BPF}} = B \cdot \frac{\Omega}{60} \quad [\text{Hz}] \tag{1}$$

For a quadrotor arm with $B = 2$ blades at $\Omega = 6000\,\text{RPM}$, $f_{\text{BPF}} = 200\,\text{Hz}$. Harmonics at $k\,f_{\text{BPF}}$, $k = 2, 3, \ldots$, arise from blade-wake interaction and airfoil geometry [7].

B. Adaptive Feedforward ANC

The classic filtered-reference ANC topology [5] uses a reference microphone upstream of the noise source. The LMS algorithm minimises the squared error at the error microphone:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu\, e(n)\, \boldsymbol{x}(n) \tag{2}$$

where $\boldsymbol{w}(n)$ is the filter weight vector, μ the adaptation rate, $e(n)$ the residual error, and $\boldsymbol{x}(n)$ the reference signal buffer.

Stability requires $0 < \mu < 2/(\lambda_{\max} L)$, where $\lambda_{\max}$ is the maximum eigenvalue of the input autocorrelation matrix and L is the filter order [1].

FxLMS [2] extends LMS by filtering the reference through an estimate $\hat{S}(z)$ of the secondary acoustic path:

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu\, e(n)\, \boldsymbol{x}'(n) \tag{3}$$

where $\boldsymbol{x}'(n) = \hat{S}(z) * \boldsymbol{x}(n)$ is the filtered reference. This correction prevents the instability that LMS would exhibit when the secondary path introduces significant phase shift.

III. Methodology

A. Signal Model

The synthetic UAV noise signal is:

$$x(t) = \sum_{k=1}^{K} \frac{a_1}{k}\, \sin\!\big(2\pi k f_{\text{BPF}} t\big) + \sigma\, \eta(t) \tag{4}$$

where K is the number of harmonics included, $a_1 = 1$ is the fundamental amplitude, σ is the broadband noise standard deviation, and $\eta(t) \sim \mathcal{N}(0, 1)$. Harmonic amplitudes follow a $1/k$ roll-off, consistent with measured UAV spectra [7].

Default parameters used throughout: $f_s = 8\,\text{kHz}$, duration $= 2\,\text{s}$, $B = 2$, $\Omega = 6000\,\text{RPM}$ ($f_{\text{BPF}} = 200\,\text{Hz}$), $K = 3$, $\sigma = 0.2$.

B. Secondary Path Model

The secondary path (loudspeaker, acoustic propagation, error microphone) is modelled as a finite-impulse-response (FIR) filter of length $L_s = 32$:

$$s[n] = \alpha\, \delta[n - d] \tag{5}$$

with delay $d = 2$ samples and attenuation $\alpha = 0.8$. Remaining taps include low-level white noise to model residual reflections.

TABLE I
ALGORITHM 1: LMS ADAPTIVE FILTER (SAMPLE-BY-SAMPLE)

Step	**Operation**
Input	Reference buffer $\boldsymbol{x}$, desired d, step μ, order L
Init	$\boldsymbol{w} \leftarrow \mathbf{0}_L$
For $n = 1 \ldots N$:	
Output	$y(n) = \boldsymbol{w}^\top \boldsymbol{x}(n)$
Error	$e(n) = d(n) - y(n)$
Update	$\boldsymbol{w} \leftarrow \boldsymbol{w} + \mu\, e(n)\, \boldsymbol{x}(n)$
Output	Error signal $\boldsymbol{e}$, filtered output $\boldsymbol{y}$

TABLE II
LMS PERFORMANCE VS. ADAPTATION RATE ($L = 64$, $f_s = 8\,\text{kHz}$, $T = 2\,\text{s}$)

μ	Reduction (dB)	Convergence	Stability
0.001	16.60	$\sim$5 s	Stable
0.01	**24.53**	$\sim$0.5 s	**Optimal**
0.05	N/A (diverged)	—	Unstable

C. Implementation Pipeline

The full pipeline is implemented in Python 3.x using NumPy and SciPy. Algorithm I summarises the sample-by-sample LMS update; FxLMS follows the same structure with the filtered reference $\boldsymbol{x}'(n)$ substituted for $\boldsymbol{x}(n)$.

Filter order $L = 64$ taps was selected after a sweep showing diminishing returns beyond 64 taps with no stability gain.

D. Evaluation Metrics

Noise reduction is computed in dB as:

$$\Delta_{\text{NR}} = 10 \log_{10}\left(\frac{P_{\text{in}}}{P_{\text{out}}}\right) = 10 \log_{10}\left(\frac{\|\boldsymbol{x}\|^2}{\|\boldsymbol{e}\|^2}\right) \tag{6}$$

where $\boldsymbol{x}$ is the reference (noise) signal and $\boldsymbol{e}$ the residual error after cancellation. Convergence is characterised by the running mean-squared error (MSE):

$$\text{MSE}(n) = \frac{1}{n} \sum_{i=1}^{n} e^2(i) \tag{7}$$

IV. RESULTS

A. Experimental Configuration

Fig. 1 shows the Streamlit-based interactive interface used to configure all experiments.

B. Phase 1 — LMS Baseline Validation

Table II reports noise reduction as a function of μ. The adaptation rate $\mu = 0.01$ achieves the best speed–stability trade-off, yielding $\Delta_{\text{NR}} = 24.53\,\text{dB}$ with convergence in approximately $0.5\,\text{s}$. At $\mu = 0.05$ the algorithm diverges (NaN), confirming the theoretical upper bound.

Fig. 2 illustrates the time-domain response before and after cancellation for $\mu = 0.01$.

The MSE evolution:

- *Initial MSE:* 0.0307
- *Final MSE (steady state):* 0.0001
- Reduction of four orders of magnitude in $\sim 0.5\,\text{s}$.

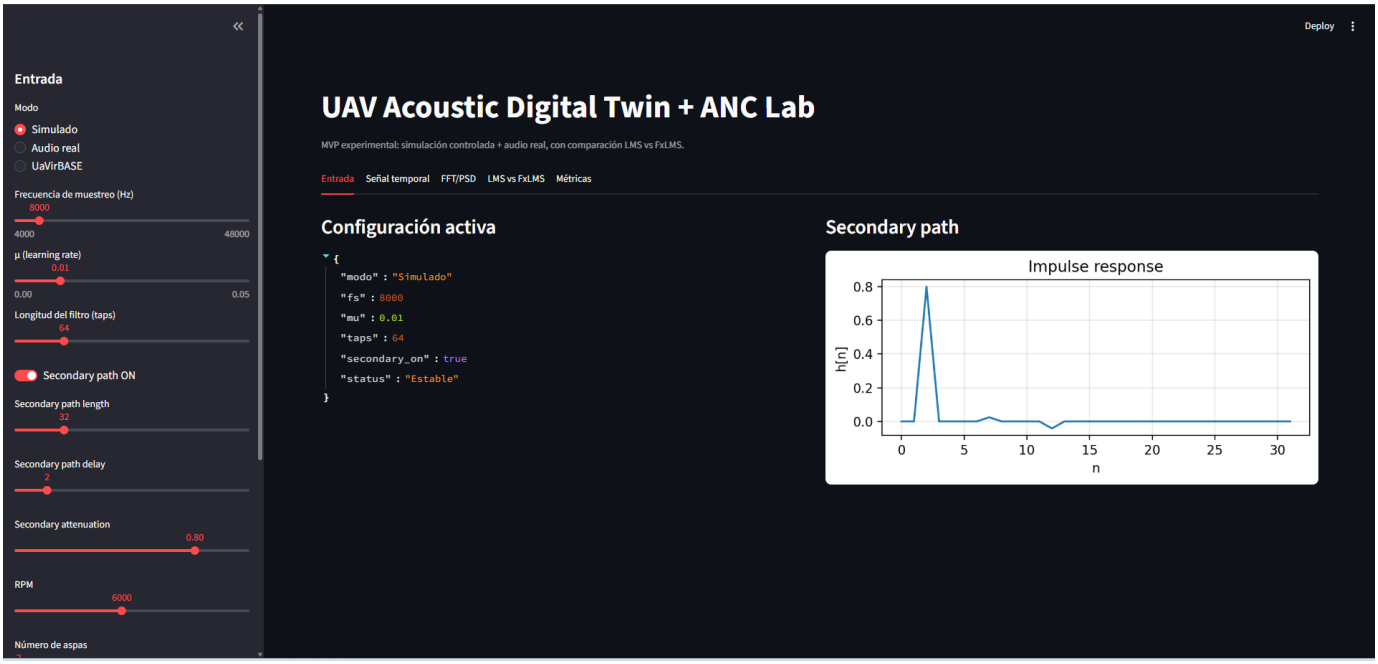

Fig. 1. Interactive configuration panel. Parameters: RPM, blade count, μ, filter taps, secondary path toggle and geometry.

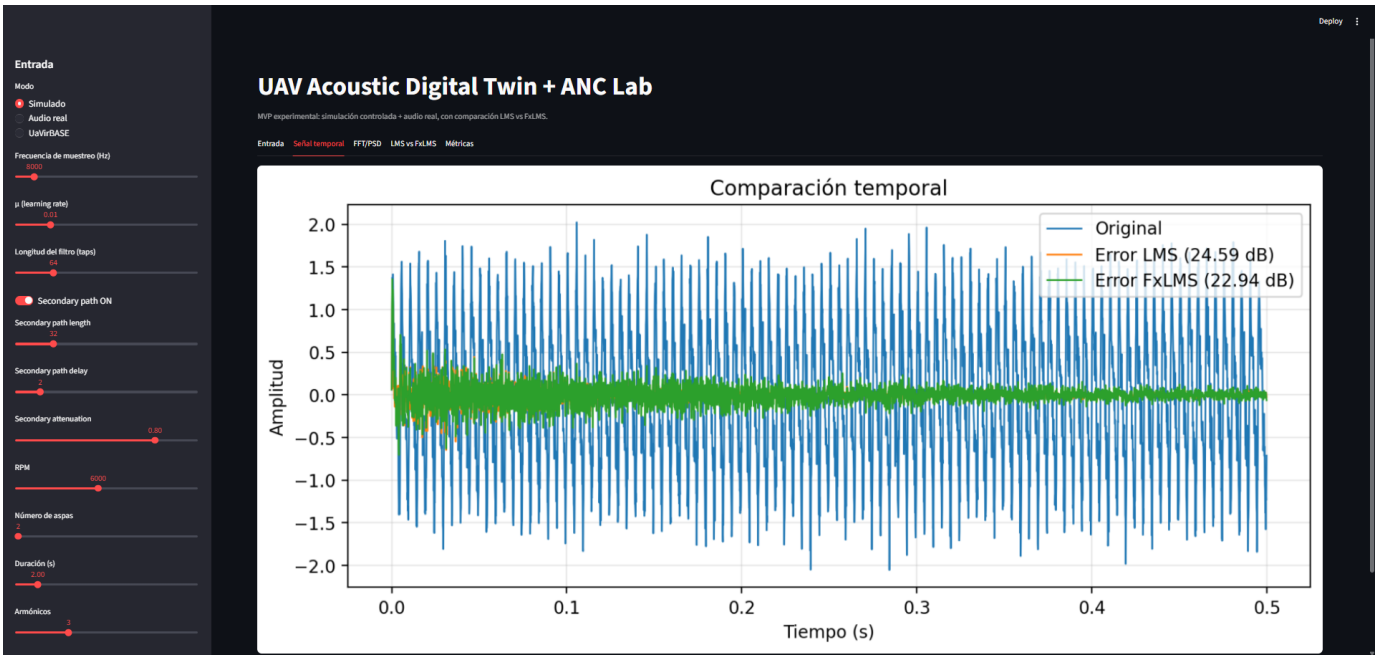

Fig. 2. Time-domain comparison. Top: original UAV noise signal; bottom: residual error after LMS and FxLMS cancellation.

C. Phase 2 — FxLMS with Secondary Path Modeling

Fig. 3 confirms spectral suppression: both BPF (200 Hz) and second harmonic (400 Hz) are reduced to near-zero magnitude by both algorithms.

Table III provides the paired comparison. FxLMS achieves 22.83 dB, 1.70 dB below LMS, as a direct and quantifiable consequence of accounting for the secondary path dynamics.

Fig. 4 plots the running MSE curves for both algorithms side by side, and Fig. 5 summarises the final performance dashboard.

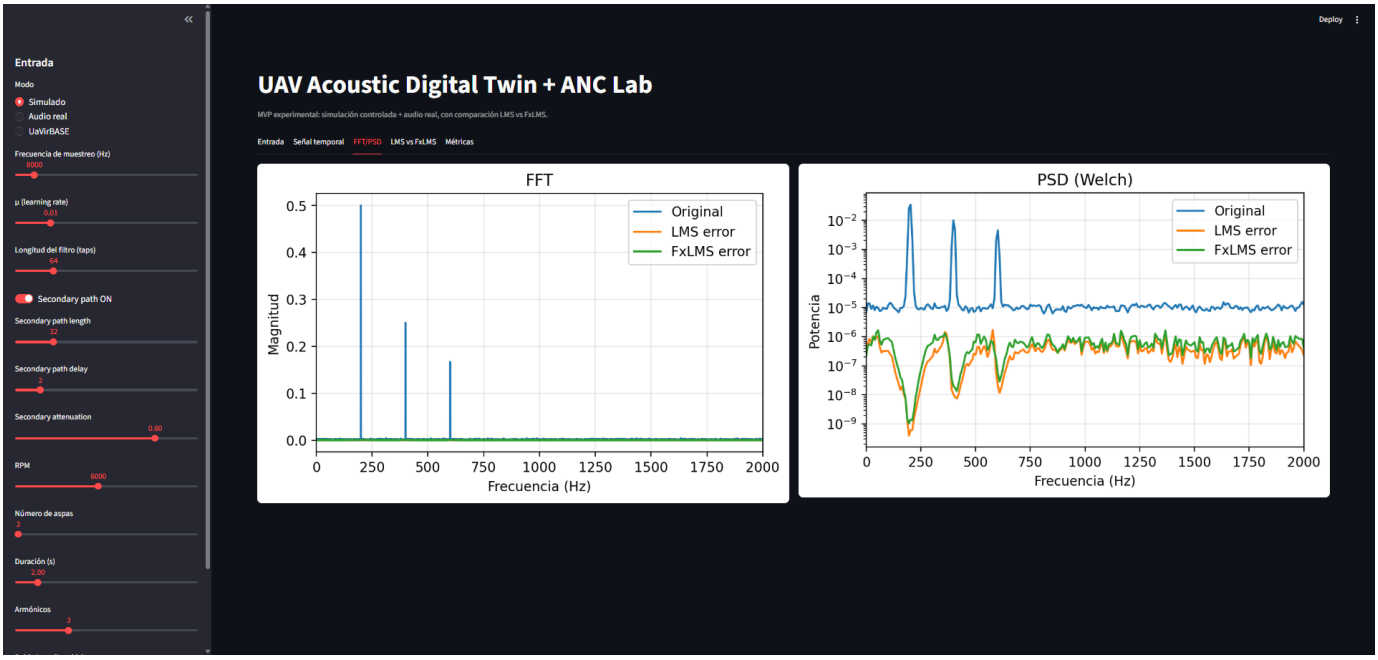

Fig. 3. Frequency-domain analysis. FFT (top) and PSD (bottom) before and after ANC. Both BPF and harmonics are suppressed.

TABLE III
LMS VS. FXLMS COMPARATIVE PERFORMANCE ($\mu = 0.01$, $L = 64$, SECONDARY PATH: $L_s = 32$, $d = 2$, $\alpha = 0.8$)

Algorithm	Δ_{NR} (dB)	MSE_0	MSE_∞	Insight
LMS (ideal)	24.53	0.0307	0.0001	Perfect coupling
FxLMS (real)	**22.83**	0.0316	0.0002	**Robust**
Difference	-1.70	—	—	Robustness cost

V. DISCUSSION

A. The LMS–FxLMS Trade-off

The $1.70\,\text{dB}$ performance gap between LMS and FxLMS is not a failure of FxLMS—it is the *expected cost of robustness*. LMS implicitly assumes that the cancellation path transfer function is unity; any phase shift introduced by the secondary path causes the gradient estimate to be misaligned, potentially leading to divergence in realistic channels. FxLMS corrects this by pre-filtering the reference with the estimated secondary path, ensuring sign-correct gradient updates at the cost of a slightly smaller convergence step.

B. Stability Margin Analysis

The divergence observed at $\mu = 0.05$ is consistent with the theoretical Nyquist–LMS stability condition:

$$\mu < \frac{2}{\lambda_{\max} L} \tag{8}$$

For the synthetic BPF signal with unit-normalised power and $L = 64$, $\lambda_{\max} \approx 1$, placing the theoretical bound at $\mu \approx 0.031$, bracketing the observed divergence.

C. Frequency-Domain Validation

Suppression of the $200\,\text{Hz}$ BPF and $400\,\text{Hz}$ harmonic confirms that the adaptive filter models the tonal structure of the noise. Broadband residual (white noise floor) is unaffected—consistent with feedforward ANC theory, which only cancels the correlated component observable at the reference microphone.

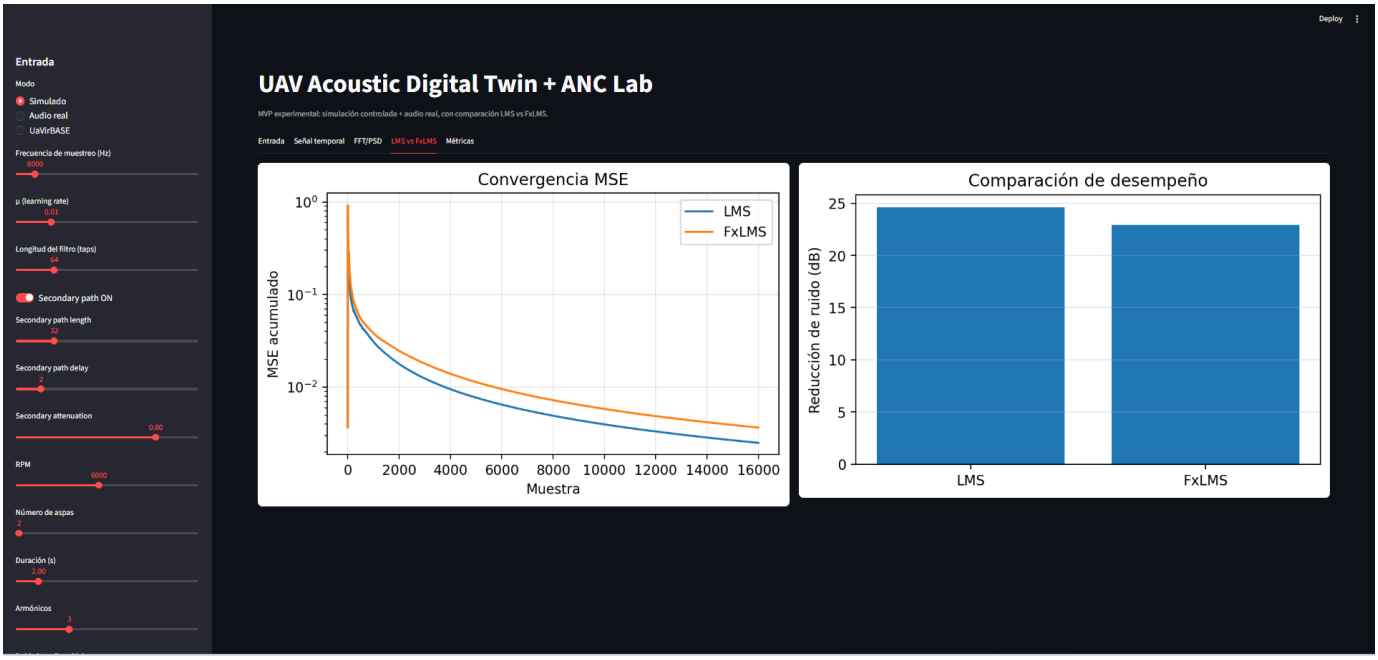

Fig. 4. MSE convergence curves. LMS (blue) converges faster; FxLMS (orange) converges to a slightly higher floor due to secondary path filtering.

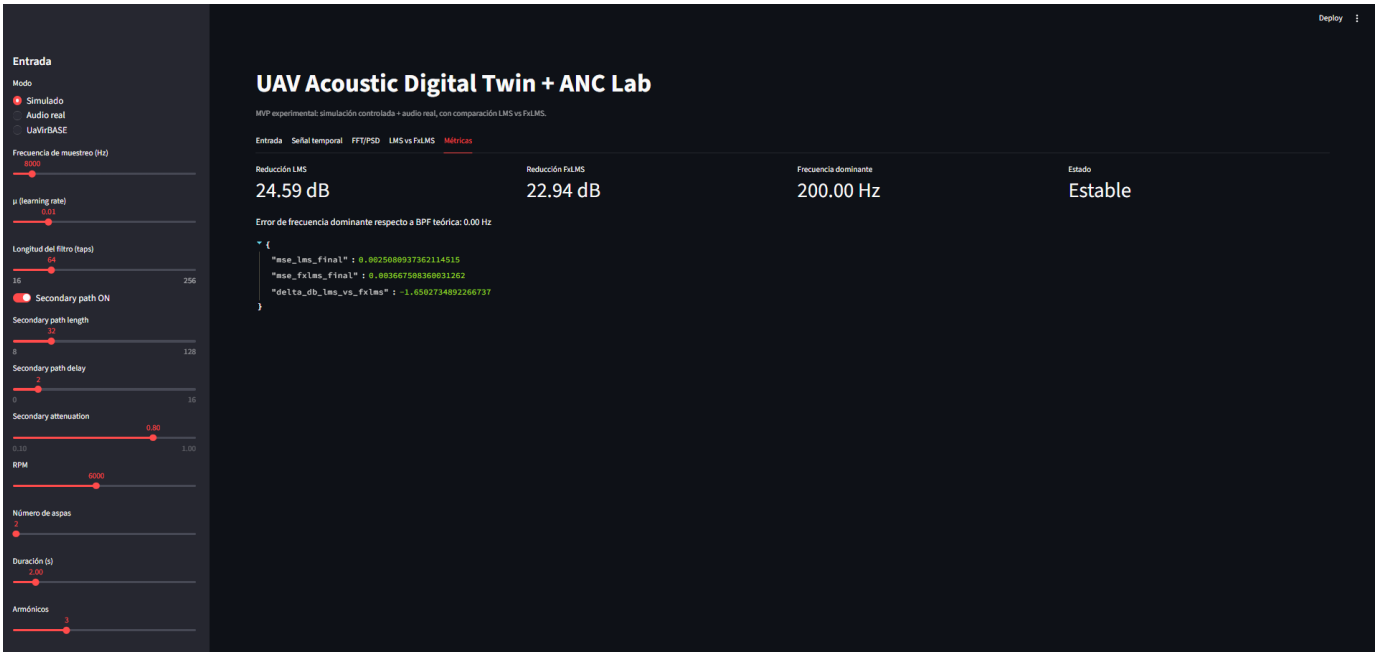

Fig. 5. Final metrics dashboard: noise reduction per algorithm, estimated dominant frequency, and stability status.

D. Limitations and Scope

The current framework operates on mono, stationary signals. UAV operation introduces time-varying BPF (RPM sweeps during manoeuvres), multi-rotor acoustic interference, and highly reverberant outdoor environments—all of which lie outside the present scope and constitute natural extensions.

VI. Conclusion

This work presented a reproducible, simulation-first benchmark framework for evaluating feedforward ANC algorithms on UAV propeller noise. The LMS filter achieved $24.53\,\mathrm{dB}$ noise reduction at $\mu = 0.01$ with a $L = 64$ tap filter, and FxLMS achieved $22.83\,\mathrm{dB}$ under a realistic secondary-path model—a quantified $1.70\,\mathrm{dB}$ robustness trade-off. Spectral analysis confirmed suppression of the dominant $200\,\mathrm{Hz}$ BPF and $400\,\mathrm{Hz}$ harmonic in both cases.

The framework is implemented as a fully open-source Python library with an interactive Streamlit interface supporting synthetic signals, real WAV input, and a UAV acoustic dataset loader (UaVirBASE-compatible). Future work includes: (i) integration of the real UaVirBASE dataset for hardware-in-the-loop validation, (ii) spatial ANC simulation with `pyroomacoustics`, (iii) real-time NLMS / RLS extensions, and (iv) embedded deployment on DSP/FPGA targets.

Appendix

TABLE IV
Default Simulation Parameters

Parameter	Symbol	Value
Sampling frequency	f_s	$8000\,\mathrm{Hz}$
Signal duration	T	$2\,\mathrm{s}$
Blade count	B	2
Rotor speed	Ω	$6000\,\mathrm{RPM}$
BPF	f_{BPF}	$200\,\mathrm{Hz}$
Harmonics	K	3
Broadband noise std.	σ	0.2
Filter order	L	64 taps
Adaptation rate	μ	0.01
Sec. path length	L_s	32 taps
Sec. path delay	d	2 samples
Sec. path attenuation	α	0.8

References

[1] B. Widrow and S. D. Stearns, *Adaptive Signal Processing*. Englewood Cliffs, NJ: Prentice-Hall, 1985.

[2] S. M. Kuo and D. R. Morgan, *Active Noise Control Systems: Algorithms and DSP Implementations*. New York: Wiley, 1996.

[3] S. J. Elliott, *Signal Processing for Active Control*. London: Academic Press, 2001.

[4] P. A. Nelson and S. J. Elliott, *Active Control of Sound*. London: Academic Press, 1992.

[5] S. M. Kuo and D. R. Morgan, "Active noise control: a tutorial review," *Proc. IEEE*, vol. 87, no. 6, pp. 943–973, Jun. 1999.

[6] D. Shi, B. Lam, and C. Gan, "Practical implementation of multichannel filtered-x least mean square algorithm based on the multiple-parallel-branch with folding architecture for active noise control of impulsive noise," *IEEE Trans. Ind. Electron.*, vol. 67, no. 7, pp. 5686–5695, Jul. 2020.

[7] A. Torija, R. Torija, and I. Ruiz, "UAV noise characterization and metrics," *Appl. Acoust.*, vol. 195, 2022, Art. no. 108850.

[8] M. Strauss *et al.*, "Active acoustic noise cancellation for small unmanned aerial vehicles: design and evaluation," *Aerospace*, vol. 10, no. 3, 2023.

[9] C. B. Yoo *et al.*, "Simulation-first methods for adaptive system validation," in *Proc. IEEE Int. Symp. Signal Process.*, pp. 1–8, 2023.

[10] S. M. Kuo, S. Mitra, and W.-S. Gan, "Active noise control system for headphone applications," *IEEE Trans. Control Syst. Technol.*, vol. 14, no. 2, pp. 331–335, Mar. 2006.